AguaClara Cornell Coagulant Dosing

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Abstract

Coagulant dosing is a critical step in the AguaClara water treatment process, involving the addition of a specific amount of coagulant to the influent water. Coagulant, a viscous chemical substance, plays a crucial role in removing various pollutants, enhancing floc formation, and neutralizing water. Currently, there exists a semi-automated coagulant dosing system which requires manual adjustment by an operator as turbidity increases. Automated coagulant dosing holds particular promise for communities with limited financial resources, as it reduces the need for a dedicated operator at all times. To move towards fully automated coagulant dosing, it is essential to develop an equation that incorporates all relevant variables affecting the coagulant dose.

The team tested the current equation using data provided from Cornell Water Filtration Plant. However, the team learned that the current dosing equation lacks consideration for floc sweeping, the process of particles moving through the floc blanket in the clarifier and colliding with big floc and joining floc to form aggregates. Thus, the objective of the Spring 2024 Coagulant Dosing team was to address this limitation by deriving a formula that accurately captures the behavior of floc sweeping in the clarifier. After understanding the physics behind the coagulant dosing equation and the primary particle's journey through the clarifier, the team accomplished their goal of finding and adding the missing variables to the equation.

Introduction

Since Cornell's Water Filtration Plant does not currently have an equation to determine how much coagulant to add, the operator decides how much to increase and decrease coagulant dosing based on the turbidity and dissolved organic matter. After testing the current equation, Equation 374 in Coagulant Automation, on CWFP's data, it was discovered that the equation overshoots coagulants at high turbidites. This leads to excess coagulant in the water which makes the formation of aggregates more difficult and affects the pH of the effluent water. In addition, excess coagulant also lowers the density of a floc. This causes the density of the floc to become too close to the density of water and as a result, flocs settle slower and are harder to capture.

$$C_{coag_{particle}} = k_{pf} C_{P_0} (C_P^{-2/3} - C_{P_0}^{-2/3}) + C_{coag_{DOM}} \tag{1}$$

After going through the derivation of the current equation, the team discovered that overshooting occurs because the current equation does not account for all floc behaviors such as floc sweeping. Floc sweeping is the process in which primary particles combine with large flocs in the clarifier to enhance flocculation and filtration. Additionally, the equation also predicts that k_{pf} should be a constant that accounts for primary particle properties and the flocculator design. k_{pf} is a variable used to represent and quantify the behavior of primary particles in the water treatment process. However, upon fitting this equation to CWFT's data suggests k_{pf} does not remain constant. Instead, k_{pf} increased exponentially as raw turbidity decreased. Overall, these findings suggest the equation is an incomplete model of coagulant dosing.

Floc sweeping takes place in the clarifier and is even more prevalent at high turbidites. In addition, the behavior of floc sweeping is dependent on the height of the floc blanket, the concentration of primary particles, the radius of primary particles, the concentration of large flocs, the radius of large flocs, the attachment efficiency, relative velocity, the number of successful collisions, the number of large flocs, the projected area of large flocs in the clarifier, and the total cross sectional area of the clarifier. Since these variables play an important role in the floc sweeping process, the team needed to take into account all these variables when deriving the new equation for coagulant when exploring the physics.

Literature Review & Previous Work

In a study titled the “Effect of Coagulation Conditions on Membrane Filtration Characteristics in Coagulation−Microfiltration Process for Water Treatment,” researchers were investigating the coagulation and water filtration process in clarification. Scientists at Seoul National University examined how the floc sweeping process affected floc formation and water quality. The researchers observed that the specific resistance of coagulated suspension varied based on floc sweeping conditions. Further, they noted that a lower resistance was observed under certain conditions which ultimately led to the formation of less compressible but more porous large flocs. This variation significantly impacted floc formation, particularly where charge neutralization demonstrated superior performance. However, under these same conditions, steady-state flux remained consistent regardless of floc sweeping. Researchers attributed this occurrence to reduced particle deposition on the floc membrane. This experiment also assessed water quality by examining the removal of natural dissolved organic matter, providing a comprehensive understanding of the interplay between the floc sweeping process, floc membrane permeability, and water treatment efficiency. Based on the research conducted at Seoul National University, a coagulant dosing equation that accounts for floc sweeping can be implemented to administer correct concentrations of coagulant. The equation should take into account factors that ensure the optimization of floc membrane permeability and water treatment efficiency by incorporating variables that influence the specific resistance of coagulated suspension and the formation of successful collisions that can form large flocs.

“Properties of flocs produced by water treatment coagulants,” investigated the effectiveness of coagulants, particularly polyaluminium chloride (PACl), compared to traditional additives like alum in water treatment processes. While the action of coagulants is typically discussed in terms of charge neutralization, the importance of precipitation and floc sweeping has not been thoroughly considered. The research compared the action of alum and three commercial PACl products on model clay suspensions using the conventional jar test procedure to monitor floc formation, floc sweeping, and breaking. Results showed that PACl products formed larger and stronger flocs than alum, with floc breakage and sweeping having significant impacts for all coagulants. The study’s preliminary results reveal that as long as floc sweeping is considered when determining the coagulant dose, PACl has the potential to be an effective coagulant in water treatment. By highlighting the importance of floc sweeping, the study underscores the necessity to account for this process in coagulant dosing. The researchers findings emphasize the need to incorporate floc sweeping variables into pre-existing coagulant dosing equations to achieve a comprehensive understanding of the water filtration process.

Kevin Sarmiento’s Thesis, “Particle Removal in Floc Blanket Clarifiers via Internal Flow Through Porous Fractal Aggregates,” provides a deeper understanding of how particles and flocs behave in the clarifier. Particles are colloidal and do not readily settle. The addition of coagulant and shear facilitates particle aggregation. The thesis hypothesizes that primary particles attach to the inside of a floc and as those flocs are captured, the capacity decreases. Particle removal essentially depends on flocculation, sedimentation, and entrapment. Kevin Sarmiento’s insight and conclusions suggest that particles are unlikely to be captured on the exterior surface of flocs in the floc blanket. This revelation introduces two critical variables for consideration: particle capture and terminal velocity.

Kim Yang also worked on coagulant dosing on Plantita and provided his experimental approach to uncover Plantita's response to kpf changes, dose changes, and DOM changes. Yang concluded that the relationship between raw turbidity and coagulant dosing during rainy days needed to be further explored to fully understand how the coagulant dose equation comes together and what could be missing. The data from his work along with CWFP's data helped in understanding how coagulant doses varied at different turbidities.

The coagulant dose is dependent on the concentration of the raw particles, measured by the raw turbidity, and the concentration of the flocculated particles, measured by the settled turbidity. Chapter 12 of the AguaClara textbook, "The Physics of Water Treatment Design," by Monroe Weber-Shirk and other collaborators, explores this relationship, see equation (1) above. The coagulant dose is also dependent on attachment efficiency α, the surface area of particles in the raw water, and a constant, k_{pf}, that is made up of the Camp-Stein parameter, $G\theta$, and the density of primary particles. This parameter defines the mean velocity gradient, G, and the residence time, θ, which is the volume of the clarifier divided by the flow rate through the tank.

To understand the behavior of flocs, Chapter 7 of the AguaClara textbook was used. For a collision to occur, the volume cleared, which is known as the space that a particle clears out as it moves through a crowd of other particles, must equal the volume occupied by one particle. This volume is dependent on the time it takes for the collision to occur and the relative velocity between particles. This dependence leads to the rate of the number of successful collisions over time. This rate increases as the concentration of primary particles and attachment efficiency increases. The rate is also indirectly proportional to the mass of a primary particle. The mass of a primary particle is equal to the density of a primary particle times the average volume per primary particle. Moreover, the rate is also related to the projected area of large flocs in the clarifier. The projected area of large flocs is only equal to the total cross-sectional area of the clarifier at high turbidites. Otherwise, the projected area is an exponential of the number of large flocs. The number of large flocs is defined by the height of the floc blanket, concentration of large flocs, and the cross-sectional area of the clarifier divided by the large floc's mass. This physics supported the additions to create the new coagulant dosing equation to be:

(2)

$$C_{coag_{particle}} = \frac{k_{pf}}{\theta_{flocculator}} C_{P,raw} \left(C_{P,clarified}{}^{-2/3} exp\left(- k_{fs} A_T \theta_{clarifier} \frac{C_{coag_{particle}}}{C_{P,raw}}\right) - C_{P,raw}{}^{-2/3} \right)$$

where

(3)

$$k_{pf} = \frac{3}{2\pi k k' \bar{G}} \left(\frac{\rho_P \pi}{6}\right)^{2/3}$$

and

(4)

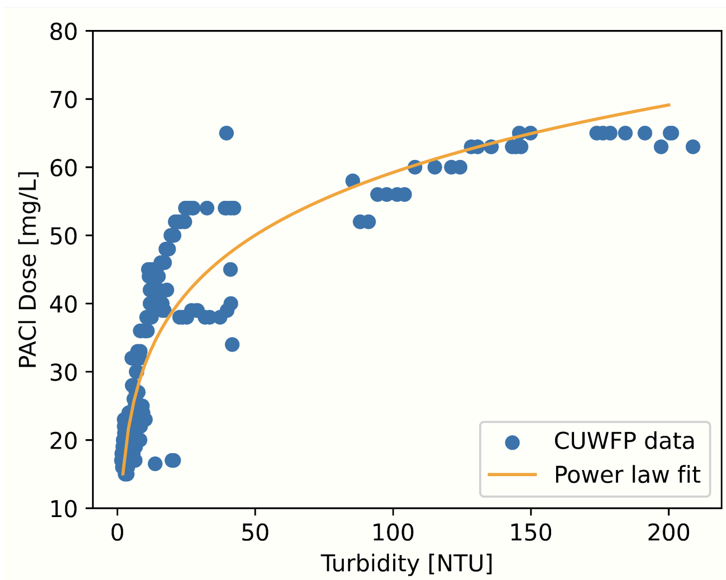
$$k_{fs} = \frac{-2k_1 k_2 v_r}{3\rho_P\left(\frac{4}{3}\pi r^3\right)}$$

Previous Coagulant Dose Response and Plantita subteams shared a common goal of lowering operational costs and understanding the interactions between coagulant and primary particles. The Coagulant Dose Response subteam determined a minimum coagulant dose and observed the effects of high coagulant dosage on effluent turbidity in spring 2019. They found that too much coagulant leads to poor floc formation, and coagulant sticks to itself more than the particles. This progress was the foundation of the coagulant dose work done since then and the variables that are being defined now. The Plantita subteam created a new coagulant dosing equation that fit a power law equation in fall 2023. The equation read,

$$C_{coag} = 81.6T^{0.104} - 72.7 \quad (5)$$

where T was influent turbidity. When graphed with the data from Cornell's Water Filtration Plant, the equation represented a line of best fit of the equation as shown below. However, the Plantita subteam believed there were other variables that played a role in the coagulant dose equation. The curve fitting done by the Plantita subteam helped to understand the data CWFP provided and move forward towards discovering the other variables present in the coagulant dose equation.

Figure 1. Plantita's Power Law Equation on a Coagulant Dose vs. Turbidity Graph.



While Plantita has a clarifier, the Cornell Water Filtration Plant has a large sedimentation tank. Fortunately, floc sweeping still takes place in the sedimentation tank. The one problem, however, is that CWFP's settling tank is much larger than Plantita's. CWFP's data can not accurately explain the cross-sectional area of the clarifier in the coagulant dose equation based off of Agua Clara's clarifier. This is why implementing the finished model on Plantita is another goal to see the true behavior.

Methods

Vertical Flow Clarifier Model

To create a new equation that accounts for floc sweeping in a vertical clarifier, the clarifier derivations were reassessed. A bottom up derivation approach was taken in determining the floc sweeping processes in a vertical clarifier system. The full derivation can be found in Appendix I, and produces the above equation (2).

(6)

$$C_{P,clarified} = C_{P,flocculated} exp(- k_{fs} A_T \theta_{floc\,filter} \frac{C_{coag_{particle}}}{C_{P,raw}})$$

As a quick overview, equation (2) was derived by combining the AguaClara textbook equation (372) with equation (6), a derived floc sweeping equation. This floc sweeping equation describes the relationship between the change in turbidity in the clarifier and the initial coagulant dosing concentration.

(7)

$$\frac{dC_P}{dt} = \frac{-k_1 \alpha C_P}{\rho_P(\frac{4}{3}\pi r^3)} v_r A_T$$

Additionally, equation (6) is the result of integrating equation (7), a differential equation describing the change in concentration of primary particles vs time due to floc sweeping.

CWFP Horizontal Flow Clarifier Model

Unlike AguaClara plants, which use vertical flow clarifiers, CWFP uses a horizontal flow clarifier. In CWFP's clarifier, flocs settle downwards as water flows horizontally. While flocs settle, they collide with—and capture—primary particles in a process called floc sweeping.

CWFP's clarifier does not have a floc filter, which greatly simplifies the clarification model. Because there is no floc filter, the residence time of flocs in the clarifier is shorter. As a result, this model assumes floc saturation/exhaustion is negligible. The model additionally assumes primary particles are uniformly distributed in the clarifier.

This derivation follows flocs and primary particles in a vertical section of the clarifier with length *Δx* as it travels through the clarifier. Primary particle capture during a time interval *dt* is then integrated through the clarifier, resulting in the predicted primary particle concentration after clarification. The second iteration of the model included dissolved organic matter (DOM) by applying Du et al.'s theory on how DOM affects coagulant dosage. Their findings suggest that coagulant only becomes available to primary particles and flocs after a fraction of the DOM coats coagulant particles. Therefore the effective coagulant level can be modeled as the initial coagulant dosage minus a fraction of the initial DOM concentration. The derivation results in the following equations with parameters γ, λ, and k_{pf}:

(8)

$$C_{primaryClarified} = C_{primary}(0)e^{-\gamma[C_{coagActual} - \lambda C_{DOM}][C_{raw} - C_{primary}(0)]/C_{raw}}$$

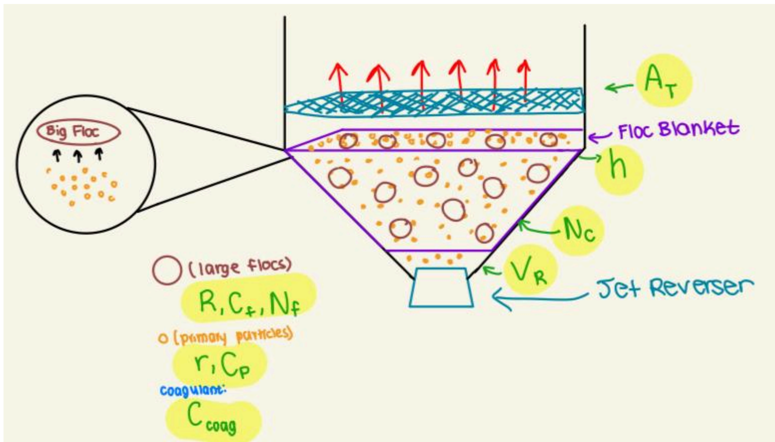
(9)

$$C_{primary}(0) = [\frac{C_{coag} - \lambda C_{DOM}}{k_{pf}C_{raw}} + C_{raw}^{-2/3}]^{-3/2}$$

The derivation of the equation describing horizontal flow clarifier floc sweeping is further detailed in Appendix II.

Additionally, MATLAB's "nlinfit" and "fit" functions from the Curve Fitting and Statistics and Machine Learning toolboxes were used to fit the CWFP horizontal flow clarifier model to data collected at CWFP from May-June 2023 (1156 data points, 30 min data interval).

Figure 2. Floc Sweeping Behavior in the Bottom of the Clarifier



Results/Analysis

Vertical Flow Clarifier Model

The team has been unable to test equation 2 (which describes vertical flow clarifier floc sweeping) given the lack of vertical flow clarifier turbidity and coagulant dosing data. But, the team has reason to assume that equation (2) is over-simplified in collision processes in the floc filter. Specifically, equation (2) assumes that each primary particle either collides or does not collide in the floc filter, however these particles have the potential to collide multiple times while passing through the floc filter. Multiple collisions between particles and flocs in the floc filter is actually highly likely due to the ratio between floc size, and floc filter height. Given that the floc filter is many times taller than the diameter of a floc, the floc filter can be approximated to many, thinner floc filters of height equal to the diameter of a floc. When passing through every single one of these floc filter cross sections, primary particles have a relatively high likelihood of a collision. Therefore, it is extremely likely that each primary particle that passes through the floc filter will collide multiple times with multiple individual flocs. Future work for correcting equation (2) for this over-simplification of collision processes would include replacing α in k_{fs} with an equation that accounts for multiple primary particle collisions.

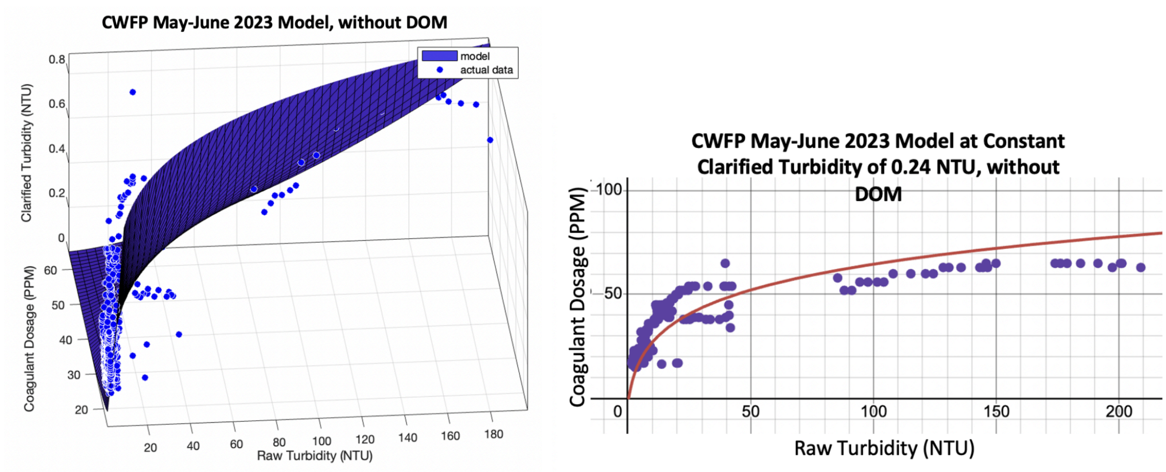
$$1 - (1 - \alpha)^{N_{collisions}} \tag{10}$$

As can be seen in equation (10), α could potentially be replaced with an equation which accounts for the increased probability of a successful collision of a primary particle due to $N_{collisions}$ number of collisions.

CWFP Horizontal Flow Clarifier Model

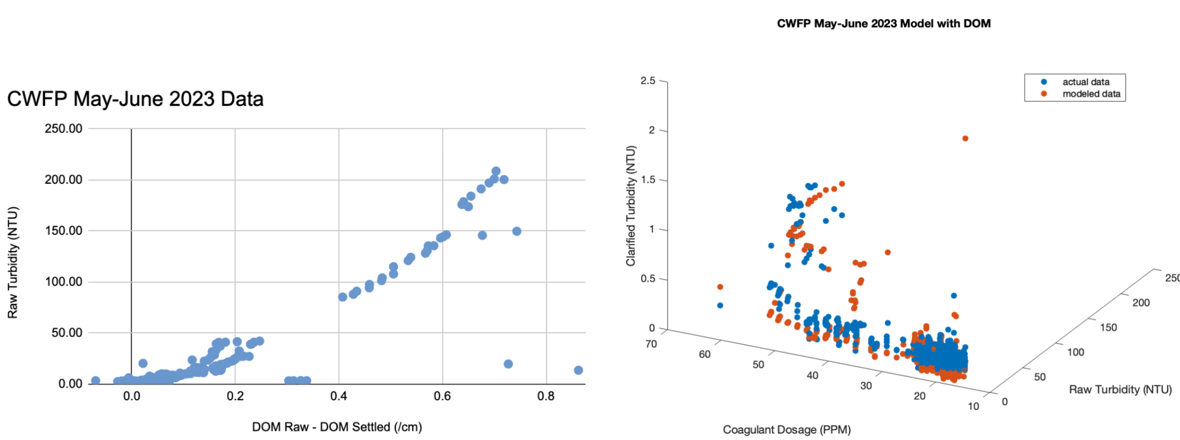
Initially dissolved organic matter (DOM) was excluded from the equation, resulting in the following parameter values (as shown in Figure 3): $k_{pf} = 3.888$ and $\gamma = 0.06464$. The root mean square error of the fit was 0.166 NTU.

Figure 3. Model fitting to CWFP data collected in May-June 2023, assuming DOM is not present ($\gamma = 0$).



Adding γ as an additional non-zero parameter to the model and using CWFP's DOM data resulted in the following parameter values (as shown in Figure 4): $k_{pf} = 3.2426$, $\gamma = 0.0214$, and $\lambda = -107.6578$. The root mean square error of the fit was 0.142 NTU. Figure 4 shows a positive, non-orthogonal relationship between DOM and raw turbidity, suggesting it is possible that UV254 measurements (ultraviolet light of wavelength 254 nm that is used to measure DOM) are influenced by clay particles and are not solely measuring DOM. This confounding effect of inorganic particles on UV254 measurements may explain the unexpected negative result for λ, as the UV254 data may have been an inaccurate representation of DOM concentration.

Figure 4. Model fitting to CWFP data collected in May-June 2023, including DOM data.



Conclusions

The model of CWFP's data (equation 9) results in an appropriate fitting of the data, as can be seen in figure 3. This reveals that the current horizontal clarifier floc sweeping model accurately predicts coagulant dosing and accounts for both flocculation and clarification processes. Given that both horizontal and vertical clarifier models are derived using very similar processes, it is likely that the vertical clarifier floc sweeping model (equation 2) is also semi-accurate in predicting proper coagulant dosing. Additionally, the model of vertical clarifier floc sweeping (equation 2) provides a starting point for modeling AguaClara clarifiers, bringing the team closer to an automated coagulant dosing system.

Future Work

Moving forward, the focus will be on several key directions. Firstly, Plantita will be run, and data will be collected using ProCoDA. This data will be utilized to verify the validity of the vertical clarifier floc sweeping model (equation 2), and solve for relevant physical constants k_{pf} and k_{fs}. Before doing so, the team needs to situate the coagulant dosing Golander pump in the CWFP plantita setup. Furthermore, data should be collected through various turbidity events to ensure global accuracy of the model. Overall, these future steps will help validate assumptions made from the horizontal clarifier floc sweeping model. Finally, the long term goal is to implement an automated coagulant dosing system in Puerto Rico.

Additionally, the team suspects that the current vertical clarifier floc sweeping model (equation 2) does not accurately describe collision mechanics between primary particles and flocs in the floc filter. Future work should include the addition of multiple collision processes in the floc filter; an example of this change can be found in equation 10.

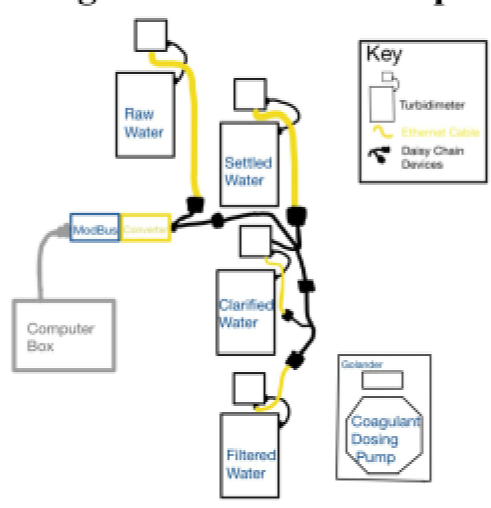
And, given that DOM is an important aspect to the coagulant dosing process, future model derivations should focus around incorporating the influence of DOM on coagulant dosing concentrations. Specifically, the results of previous applications of DOM in the CWFP model indicate that the model is not accurately representing the influence of DOM. The team suspects that current methods of measuring DOM (using UV254) are not accurate. Future work could include reassessment of DOM measurement processes, and research into the influence of clay particles (turbidity) on UV254 readings.

Manual

Modbus Setup

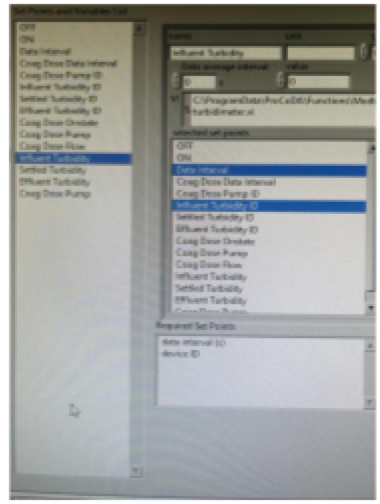
Through daisy chain devices, the Modbus attaches to a RS 485 converter and that converter connects to the four turbidimeters. Ethernet cables run from the turbidimeter to the daisy chain device, and that device connects to the converter. The Modbus setup ensures that all four turbidimeters are connected to the computer box.

Figure 5. ModBus Setup



To get ProCoDA communicating with the turbidimeters, select “Edit Rules” on the Configuration tab. Add a set point for the data interval and set a time interval of 5 seconds. From there, add a second set point for the turbidimeter address. Name the turbidimeter, “Type of Turbidimeter ID,” keep the type as “Constant,” and set the value equal to the unit ID. Next, add a third setpoint that will be the measured turbidity. The measured turbidity will be a variable dependent on the data interval and its ID. For example, for influent turbidity, the data interval and the influent turbidity ID must be selected as shown in Figure 7. After, ProCoDA should read the value on the influent turbidimeter and display the turbidity in the value box. If not, the errors will be displayed on the right side of the Configuration tab. See [ProCoDA documentation] to uncover unexpected errors.

Figure 6. Set Points for Influent Turbidity



An error faced during this process was being unable to get ProCoDA to display the same turbidity as the turbidimeter. After carefully looking at the manual from the Agua Clara Textbook, it was realized that wiring was done incorrectly; two wires were swapped and the middle wire was unplugged. After rewiring, the four turbidimeters connected to ProCoDA and ProCoDA outputted the turbidity displayed on the turbidimeters.

References

AguaClara Hydraulic Flocculation Model. AguaClara Hydraulic Flocculation Model - AguaClara v1.4.35 documentation. (n.d.). https://aguaclara.github.io/Textbook/Flocculation/Floc_Model.html

Clarification introduction. Clarification Introduction - AguaClara v1.4.35 documentation. (n.d.). https://aguaclara.github.io/Textbook/Clarification/Clarifier_Intro.html

Coagulant Automation. Coagulant Automation - AguaClara v1.4.35 documentation. (n.d.). https://aguaclara.github.io/Textbook/Operation/Coagulant_Automation.html

Du, Y., Pennock, W. H., Weber-Shirk, M. L., & Lion, L. W. (2019). Observations and a Geometric Explanation of Effects of Humic Acid on Flocculation. *Environmental Engineering Science*, *36*(5), 614–622. https://doi.org/10.1089/ees.2018.0405

ProCoDA: Process Control and Data Acquisition. ProCoDA: Process Control & Data Acquisition - AguaClara v1.4.35 documentation. (n.d.). https://monroews.github.io/EnvEngLabTextbook/ProCoDA/ProCoDA.html

Yao, K.-M., Habibian, M. T., & O'Melia, C. R. (1971). Water and waste water filtration. Concepts and applications. *Environmental Science & Technology*, *5*(11), 1105–1112. https://doi.org/10.1021/es60058a005

Appendix I

The vertical floc sweeping equation derivation can be found here.

Appendix II

The derivation for the CWFP model can be found here.